Summary Notes:
Administering Network Traffic in Azure

Overview

Here is an overview of the topics covered in this session.

1. Session 1
    a. System Routes
    b. User-Defined Routes
    c. Service Endpoints and Private Links
    d. VNet Peering
    e. Service Chaining

2. Session 2
    a. Azure Load Balancer
    b. Implementing the Public Load Balancer
    c. Implementing the Internal Load Balancer
    d. Session Persistence
    e. Azure Application Gateway
    f. Components for the Configuring Application Gateway

Session 1: Network Routing and Endpoints

1. System Routes

- Azure's built-in routing automatically handles traffic between:
  - **VMs within the same subnet**: Internal communication occurs within the same subnet without any additional setup
  - **VMs across different subnets in the same VNet**: Handles subnet-to-subnet communication within a virtual network
  - **VMs and the internet**: Allows VMs to communicate with external networks via Azure's routing system
  - **VNet-to-VNet VPN connections**: Supports traffic between virtual networks using VPN gateways
  - **Site-to-site VPN and ExpressRoute connections**: Manages secure tunnel-based or dedicated circuit communication between on-premises and Azure networks
- **Key benefits**: System Routes provides seamless default routing and aids in troubleshooting and optimising the traffic flow. Understanding system routes is vital for managing and extending network configurations, especially for hybrid connectivity

2. User-Defined Routes (UDRs)

- Administrators can override system routes to control traffic patterns by creating UDRs
- The applications of UDRs are as follows:
  - **Custom traffic control**: Define explicit routing rules to control which routes traffic takes within a VNet
  - Next Hop Options
    - **Virtual network gateway**: Direct traffic through a VPN gateway
    - **Virtual network**: Route to another virtual network
    - **Internet**: Send traffic to external networks
    - **Virtual appliance**: Direct traffic through a network virtual appliance (e.g., firewall)
- **Advantages**: UDRs enables targeted routing that can enhance network security and performance; for example, UDRs can enforce routing through security appliances, allowing organisations to secure and inspect network traffic

3. Service Endpoints and Private Links

- **Service endpoints**: These create secure, direct links to Azure services by keeping traffic within Azure's backbone network and bypassing the public internet
  - **Use cases**: Often used to secure Azure Storage or SQL databases by restricting access to specific VNets/subnets, reducing exposure to the public internet
  - **Implementation**: Typically takes 15 minutes to configure and significantly boosts security by ensuring traffic does not leave Azure's network
- **Private Links**: Offer private connectivity to Azure services over a private IP in a VNet
  - **Key benefits**: Maintain traffic within Azure's private network, ensuring greater privacy and security, especially for data-sensitive environments
  - **Use cases**: Are essential for compliance where data must avoid public internet exposure, providing private connectivity for Azure resources such as storage, SQL database and other services within the user's private network

4. VNet Peering

- Regional and Global Peering: Connects Azure VNets, whether within the same or across different regions, using Azure's backbone network, eliminating the need for a VPN
  - **Regional Peering**: Allows VNets within the same region to communicate
  - **Global Peering**: Supports connectivity across regions, which is useful for multi-region deployments
- Benefits
  - **Data transfer efficiency**: Traffic is routed directly within Azure's network, avoiding public exposure
  - **Security**: Keeps traffic within the Azure infrastructure, adding another layer of isolation
- **Gateway transit and connectivity**: In hub-and-spoke architectures, gateway transit enables spokes to share the central VNet's gateway, streamlining network management and reducing costs by centralising VPN or ExpressRoute connections

        - **Requirements**: VNets must have non-overlapping IP ranges to use gateway transit effectively.

5. Service Chaining
    - Service chaining directs traffic through network appliances, such as firewalls or intrusion detection systems, before it reaches its final destination. This is enabled by UDRs and is crucial for environments needing detailed traffic inspection or security controls.

Session 2: Load Balancing Solutions

1. Azure Load Balancer

- **Overview**: A Layer 4 load balancer that distributes TCP/UDP traffic across VMs or services to enhance application availability and resiliency
- Types of Load Balancers
    - Public Load Balancer: Exposes Azure resources to the internet by mapping public IP addresses to internal resources
    - Internal Load Balancer: Routes traffic within a VNet, which is ideal for backend services or applications not exposed to the internet
- Key Use Cases
    - **High availability**: Ensures continuous availability by distributing incoming traffic across multiple instances
    - **Health probes**: Uses health checks to monitor backend resources and reroute traffic to healthy instances, reducing downtime during failures
- Load Balancer SKUs
    - **Basic SKU**: Suitable for simpler applications with fewer backend instances and limited features (no availability zone support)
    - **Standard SKU**: Provides advanced features such as support for up to 1,000 instances, zonal redundancy and better diagnostics for larger, high-availability applications

2. Implementing the Public Load Balancer

- **Setup**: Involves assigning a public IP to the Load Balancer's frontend configuration and creating Load Balancer rules to map frontend IPs to backend resources

- **Health probes**: Monitors backend resources to ensure only healthy instances receive traffic, which is critical for web applications accessible to internet users

3. Implementing the Internal Load Balancer

- **Usage**: For private traffic within a VNet, such as distributing requests between tiers in a multi-tier application
- **Private IPs**: Uses internal IPs to manage traffic across private resources, optimising traffic flow between application tiers without public exposure

4. Session Persistence

- Options
  - **None (default)**: Any backend instance can handle client requests
  - **Client IP**: Traffic from the same client IP consistently goes to the same instance
  - **Client IP and Protocol**: Maintains consistency for traffic from the same client IP and protocol, which is useful for applications needing session continuity

5. Azure Application Gateway

- **Overview**: A Layer 7 load balancer for HTTP/S traffic with advanced features such as path-based routing, SSL termination and WAF
- Key Features
  - **Path-based routing**: Routes requests based on URL paths, e.g., `/images` requests can go to one backend and `/app` requests to another
  - **Multi-site routing**: Allows routing based on the domain, enabling multiple websites on the same Application Gateway
- **SSL Termination**: Offloads SSL decryption to reduce backend load, simplifying SSL management
- **Web Application Firewall (WAF)**: Provides protection against vulnerabilities such as SQL Injection and XSS; configurable with custom rules to handle specific security requirements

- **Health Probes**: Similar to the Load Balancer, these probes check the health of backend resources, ensuring traffic only reaches responsive servers

6. Components for the Configuring Application Gateway

- **Frontend IP**: The external IP clients use to connect to the application
- **Listeners**: Listen for incoming requests on specific ports, e.g., port 80 for HTTP
- **Routing Rules**: Define how traffic is routed to backend resources based on the listener's configurations
- **Backend Pools**: Contains backend servers such as VMs, App Services or on-premises servers that receive traffic